Assignment 3: Variational Formulations and Energy Principles

ME/BME 736 — Advanced Finite Element Methods

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2026-09-11

Show the main steps in each derivation. This assignment concludes the continuous mathematical and variational foundations developed in Chapters 2–4. No finite element discretization is required.

Problem 1 (25 marks): Applying the Lax–Milgram theorem to diffusion with a zero-order term

For help: See Chapter 3, Section *Well-posedness and stability.* For the definitions of continuity and coercivity and for the Cauchy–Schwarz and trace inequalities, see Chapter 2, Sections *Continuity, coercivity, and well-posedness* and *Reference inequalities.*

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain with

$$\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}, \qquad \Gamma_D \cap \Gamma_N = \emptyset,$$

where Γ_D has positive boundary measure. Consider

$$-\nabla\cdot(k\nabla u) + cu = f \qquad \text{in } \Omega,$$

with

$$u = 0 \qquad \text{on } \Gamma_D,$$

Define the diffusion-flux vector and its outward normal component by

$$\boldsymbol{q} = k\nabla u, \qquad q = \boldsymbol{q}\cdot\boldsymbol{n}.$$

The boundary conditions are

$$\boldsymbol{q}\cdot\boldsymbol{n} = q \qquad \text{on } \Gamma_N.$$

The additional term cu contains no derivative of u and is therefore a **zero-order term**. In diffusion models it is also commonly called a reaction term.

Assume

$$k, c \in L^\infty(\Omega), \qquad 0 < k_0 \le k(\boldsymbol{x}) \le k_1, \qquad 0 < c_0 \le c(\boldsymbol{x}) \le c_1$$

almost everywhere in Ω, with

$$f \in L^2(\Omega), \qquad q \in L^2(\Gamma_N).$$

Use

$$V = H^1_{0,\Gamma_D}(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

1. Starting from the strong problem, derive

   $$a(u, v) = \ell(v) \qquad \forall v \in V,$$

   and identify $a(w, v)$ and $\ell(v)$.
2. You may use the estimates

   $$|a(w, v)| \le (k_1 + c_1) \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)},$$

   $$a(v, v) \ge \min\{k_0, c_0\} \|v\|^2_{H^1(\Omega)},$$

   and

   $$|\ell(v)| \le \left(\|f\|_{L^2(\Omega)} + C_{\text{tr}} \|q\|_{L^2(\Gamma_N)}\right) \|v\|_{H^1(\Omega)}.$$

   Explain which hypotheses of the Lax–Milgram theorem these estimates establish and state the conclusion of the theorem for this problem.

Problem 2 (20 marks): From an energy functional to the variational problem

For help: See Chapter 3, Section *Energy minimization and first variation.* For directional derivatives and first variations, see Chapter 2, Section *Directional derivatives and first variations.*

Use the same coefficient functions k and c, source f, normal datum $q = \boldsymbol{q} \cdot \boldsymbol{n}$, and boundary partition as in Problem 1. Now allow a prescribed value

$$u = g \qquad \text{on } \Gamma_D,$$

and define

$$\mathcal{U}_g = \{z \in H^1(\Omega) : z = g \text{ on } \Gamma_D\}, \qquad \mathcal{V}_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

For $z \in \mathcal{U}_g$, consider

$$\Pi(z) = \frac{1}{2}\int_\Omega (k|\nabla z|^2 + cz^2)\ \mathrm{d}x - \int_\Omega fz\ \mathrm{d}x - \int_{\Gamma_N} qz\ \mathrm{d}s.$$

1. Let $u \in \mathcal{U}_g$. A variation, or perturbation, of u should preserve the prescribed value g on Γ_D. Explain why choosing $v \in \mathcal{V}_0$ ensures that

   $$u + \varepsilon v \in \mathcal{U}_g$$

   for every scalar ε.
2. Compute the first variation

   $$\delta\Pi(u; v) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Pi(u + \varepsilon v)\Big|_{\varepsilon=0}.$$

3. Set

   $$\delta\Pi(u; v) = 0 \qquad \forall v \in \mathcal{V}_0.$$

   Show that the resulting equation can be written as

   $$a(u, v) = \ell(v) \qquad \forall v \in \mathcal{V}_0,$$

   where the bilinear form a and linear functional ℓ are exactly the same forms identified in Problem 1.

Problem 3 (30 marks): From strong form to variational form in linear elasticity

For help: Review Chapter 3, Section *Canonical scalar diffusion problem*, especially *Weighted residual and integration by parts*, *Trial and test spaces*, and *Variational problem.* For the corresponding elasticity model, see Chapter 4, Section *Small-strain elasticity.*

Let an elastic body occupy a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$, where $d = 2$ or 3. Its boundary is divided into two non-overlapping parts,

$$\partial\Omega = \overline{\Gamma_u} \cup \overline{\Gamma_t}, \qquad \Gamma_u \cap \Gamma_t = \emptyset.$$

The unknown displacement field is

$$\boldsymbol{u} : \Omega \to \mathbb{R}^d.$$

The quasistatic equilibrium equation is

$$-\nabla\cdot\boldsymbol{\sigma} = \rho\boldsymbol{b} \qquad \text{in } \Omega,$$

with boundary conditions

$$\boldsymbol{u} = \overline{\boldsymbol{u}} \qquad \text{on } \Gamma_u,$$

and

$$\boldsymbol{\sigma}\boldsymbol{n} = \bar{\boldsymbol{t}} \qquad \text{on } \Gamma_t.$$

Here:

- $\boldsymbol{u}$ is the displacement field;
- $\overline{\boldsymbol{u}}$ is the prescribed displacement on Γ_u;
- $\boldsymbol{\sigma}$ is the Cauchy stress tensor;
- $\boldsymbol{n}$ is the outward unit normal to the boundary;
- $\bar{\boldsymbol{t}}$ is the prescribed traction vector, or force per unit area, on Γ_t;
- ρ is the mass density;
- $\boldsymbol{b}$ is the body force per unit mass, such as gravitational acceleration;
- therefore $\rho\boldsymbol{b}$ is the body force per unit volume.

For small deformations, the infinitesimal strain tensor is

$$\boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2}\left(\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\right).$$

For an isotropic linear elastic material,

$$\boldsymbol{\sigma}(\boldsymbol{u}) = 2\mu\boldsymbol{\varepsilon}(\boldsymbol{u}) + \lambda(\nabla\cdot\boldsymbol{u})\boldsymbol{I},$$

where λ and μ are the Lamé material parameters and $\boldsymbol{I}$ is the identity tensor. The stress tensor $\boldsymbol{\sigma}$ is symmetric.

Begin with a smooth vector test function $\boldsymbol{v}$ satisfying

$$\boldsymbol{v} = 0 \qquad \text{on } \Gamma_u.$$

You may use the tensor integration-by-parts identity

$$\int_\Omega (-\nabla\cdot\boldsymbol{\sigma})\cdot\boldsymbol{v}\ \mathrm{d}x = \int_\Omega \boldsymbol{\sigma} : \nabla\boldsymbol{v}\ \mathrm{d}x - \int_{\partial\Omega} (\boldsymbol{\sigma}\boldsymbol{n})\cdot\boldsymbol{v}\ \mathrm{d}s.$$

Because $\boldsymbol{\sigma}$ is symmetric,

$$\boldsymbol{\sigma} : \nabla\boldsymbol{v} = \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\boldsymbol{v}),$$

and for isotropic linear elasticity,

$$\boldsymbol{\sigma}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) = 2\mu\,\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) + \lambda\,(\nabla\cdot\boldsymbol{u})(\nabla\cdot\boldsymbol{v}).$$

1. Multiply the equilibrium equation by $\boldsymbol{v}$, integrate over Ω, and use the identities above together with the boundary conditions to obtain

$$\int_\Omega \left[2\mu\,\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) + \lambda(\nabla\cdot\boldsymbol{u})(\nabla\cdot\boldsymbol{v})\right]\mathrm{d}x = \int_\Omega \rho\boldsymbol{b}\cdot\boldsymbol{v}\ \mathrm{d}x + \int_{\Gamma_t} \bar{\boldsymbol{t}}\cdot\boldsymbol{v}\ \mathrm{d}s.$$

2. Assume

$$\lambda, \mu \in L^\infty(\Omega), \qquad \rho \boldsymbol{b} \in L^2(\Omega; \mathbb{R}^d), \qquad \bar{\boldsymbol{t}} \in L^2(\Gamma_t; \mathbb{R}^d),$$

   and assume that the prescribed displacement $\overline{\boldsymbol{u}}$ is the boundary value of a function in $H^1(\Omega; \mathbb{R}^d)$. Following the same reasoning used for the scalar diffusion problem in Chapter 3, identify appropriate trial and test spaces for $\boldsymbol{u}$ and $\boldsymbol{v}$ so that all terms in the integral identity are well defined.
3. Define the bilinear form $a(\boldsymbol{u}, \boldsymbol{v})$ and the linear functional $\ell(\boldsymbol{v})$, and write the complete variational problem in the form

$$\text{find } \boldsymbol{u} \in \mathcal{U}_{\overline{\boldsymbol{u}}} \text{ such that} \qquad a(\boldsymbol{u}, \boldsymbol{v}) = \ell(\boldsymbol{v}) \qquad \forall \boldsymbol{v} \in \mathcal{V}_0.$$

Problem 4 (25 marks): A manufactured solution in linear elasticity

For help: See Chapter 4, Section *Small-strain elasticity*, especially *Infinitesimal strain*, *Stress, traction, and balance of momentum*, and *Linear elastic constitutive response*. For tensor notation and tensor-vector multiplication, see Chapter 2, Section *Scalars, vectors, and tensors*.

A manufactured solution starts from a chosen displacement field and determines the body force and boundary tractions for which that displacement is an exact solution.

Let

$$\Omega = (0, L) \times (0, H) \subset \mathbb{R}^2,$$

and consider isotropic small-strain linear elasticity with

$$\boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2} \left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T \right),$$

and

$$\boldsymbol{\sigma}(\boldsymbol{u}) = 2\mu \boldsymbol{\varepsilon}(\boldsymbol{u}) + \lambda \, \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{u})) \, \boldsymbol{I},$$

where $\lambda > 0$ and $\mu > 0$ are material constants and $\boldsymbol{I}$ is the 2×2 identity tensor.

Consider the displacement field

$$\boldsymbol{u}(x_1, x_2) = \begin{bmatrix} \alpha x_1^2 \\ 0 \end{bmatrix},$$

where α is a prescribed constant. Let the left edge $x_1 = 0$ be a displacement boundary with $\boldsymbol{u} = 0$, and treat the remaining three edges as traction boundaries.

1. Compute

$$\nabla \boldsymbol{u}, \qquad \boldsymbol{\varepsilon}(\boldsymbol{u}), \qquad \mathrm{tr}(\boldsymbol{\varepsilon}(\boldsymbol{u})), \qquad \boldsymbol{\sigma}(\boldsymbol{u}).$$

2. Compute

$$-\nabla \cdot \boldsymbol{\sigma}(\boldsymbol{u})$$

and hence determine the body-force field $\rho \boldsymbol{b}$ for which the prescribed displacement is an exact equilibrium solution.

3. Determine the outward unit normal on the right, top, and bottom edges and compute the traction

$$\bar{\boldsymbol{t}} = \boldsymbol{\sigma}(\boldsymbol{u})\boldsymbol{n}$$

on each edge.